

Auctions with Risk Reserves for Scarce AI Research

Zhenning Wang
Duke Kunshan University
Kunshan, China
zw317@duke.edu

Qianshuo (Aaron) Wang
Duke Kunshan University
Kunshan, China
qw124@duke.edu

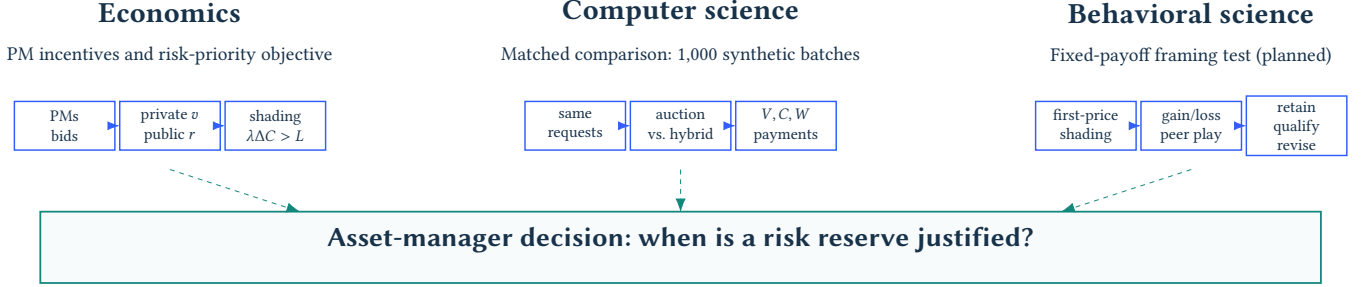


Figure 1: Evidence chain: theory defines the value–risk trade-off; 1,000 matched synthetic batches show mean hybrid gains of 9.78 at $\lambda = 1$ and -8.52 at $\lambda = 0.5$. Classroom framing evidence remains planned. The decision is whether risk coverage justifies displaced private value.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** FP8. **Symposium:** D.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Three connected research questions and verified gap

T. Rowe Price’s September 2026 expansion of Claude for portfolio managers (PMs) and analysts motivates our setting [6]. We model an active asset manager during earnings season: 20 PMs compete for four AI research slots. Scarcity and allocation rules are research assumptions, not documented T. Rowe Price practices. PM willingness to pay may neglect firm-level risk priorities, affecting PMs, risk teams, and clients.

Verified gap. Sheng et al. [3] study GenAI adoption and investment performance, not internal access rules. Stokely et al. [5] study compute-quota markets; Sivan et al. [4] already incorporate heterogeneous values and organizational priorities into dynamic ML-resource pricing. Pathak et al. [2] develop non-price reserve systems for medical rationing. Building on these dimensions, we examine a specific unresolved design choice: verified risk reserves alongside paid bidding for indivisible AI research slots, with behavioral sensitivity. This is an application-level comparison, not a claim to invent markets or reserves; medical objectives do not transfer to finance.

Q1—Economics: When does risk priority justify displaced private value?

Q2—Computation: How do reserve size and value–risk alignment change allocation, payments, utility, efficiency, and risk coverage versus pure first-price auction and FCFS?

Q3—Behavior: With payoffs fixed, does loss framing change bids enough to alter the mechanism comparison?

Shared question: Can a hybrid mechanism that combines auction-based allocation with reserved capacity for objectively high-risk research requests improve the allocation of scarce AI research capacity relative to a pure first-price auction? Economics defines improvement; computation tests its conditions; behavior probes the bidding assumptions. None alone resolves the decision. Figure 1 links these tasks; synthetic comparisons are reported below; behavioral evidence remains planned.

2 Answer to Q1: incentives and strategic model

Environment and rules. In one batch, $N = 20$ risk-neutral PMs compete for $K = 4$ identical premium AI research slots, at most one per PM. PM i privately knows $v_i \in [0, 100]$, holds $M_i = 100$ credits, and bids $b_i \in [0, 100]$. Unspent credits have fixed unit outside value; no later strategic stage is modeled. The risk team freezes public $r_i = 100E_iS_i$, with normalized exposure and severity $E_i, S_i \in [0, 1]$, before bidding. Hybrid awards one free slot to the highest-risk request with $r_i \geq r^*$; that PM exits. Remaining PMs simultaneously submit sealed bids for three slots; highest bids win and pay their own bids. If none is eligible, auction all four. Net utility is $u_i = x_i v_i - p_i$.

Game theory. Conditional on the reserve outcome, private-value bidding has Bayesian Nash equilibrium as its rational benchmark [1]. Separately, symmetric risk-neutral bidders with independent $U[0, 1]$ values in a single-slot first-price auction shade bids: $b(v) = (n - 1)v/n$ [7]. This is not asserted as the multi-slot equilibrium; simulations use explicit stylized bids.

Social choice. For allocation A , the firm evaluates $V(A) = \sum_{i \in A} v_i$, $C(A) = \sum_{i \in A} r_i$, and $W(A) = V(A) + \lambda C(A)$, $\lambda \geq 0$. Payments are internal transfers, reported separately.

Mechanism design. Compare pure first-price auction P , hybrid H , and FCFS. With $\Delta C = C(H) - C(P)$ and $L = V(P) - V(H)$, the

conditional prediction is $W(H) > W(P)$ iff $\lambda\Delta C > L$: risk coverage must justify displaced private value.

3 Answer to Q2: computation, auction comparison, and artifacts

Inputs and algorithm. With Section 2’s $N = 20, K = 4, M = 100$, draw independent $v_i \sim U[0, 100]$ and $E_i, S_i \sim U[0, 1]$; set $r_i = 100E_iS_i$, $r^* = 60$, $R = 1$, $\lambda = 1$, and assumed bids $b_i = 0.8v_i$. For each matched batch: freeze requests; allocate the free reserve to the highest eligible risk; remove that PM; auction remaining capacity by descending bid; charge own bids; stop. Unused reserves return to auction; lower PM ID breaks ties. Compare pure four-slot first-price auction and free FCFS by random exogenous arrival. Bids are held fixed, not solved as multi-slot equilibrium strategies.

Fresh-run answer. Seed 20260926 and 1,000 independent batches give the following means (internal payments P ; total PM utility U):

Rule	V	C	W	P	U
Pure	353.08	101.35	454.43	282.46	70.62
Hybrid	326.26	137.95	464.20	226.98	99.28
FCFS	199.45	100.43	299.88	0	199.45

Hybrid gains 36.59 risk units but loses 26.82 value units: mean $\Delta W = 9.78$ at $\lambda = 1$ (Monte Carlo SE 0.89), versus -8.52 at $\lambda = 0.5$. The mean break-even weight is 0.733. Under negative value–risk rank alignment, the gain becomes -3.24 at $\lambda = 1$. Thus reserve adoption depends on priorities and alignment, not eligibility alone. The repository reports winner IDs, payments, utilities, efficiency and eligible access, plus 36 sensitivity scenarios; six tests and a clean rerun passed. Appendix A.2 gives pseudocode, parameters and verification. This synthetic private-value allocation excludes common financial values and future rounds; risk coverage is not avoided loss.

GitHub: https://github.com/dku-comsci-econ206-Autumn2026/PS2_Qianshuo_Zhenning (final tested commit: 59b1ead).

Notebook: https://colab.research.google.com/drive/1I-J52wxGaAIwJK_LjtghBnXS2GxY0U2V#scrollTo=QI1TUTULTNPj.

Hugging Face: https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning.

A0 poster: <https://canva.link/vlx5ljli23z7qk8>.

4 Answer to Q3: behavior and interdisciplinary integration

Q3 is a planned human-bidding task, distinct from the completed Colab simulation. Four bidders compete for one slot in a first-price auction. With symmetric risk-neutral bidders and independent uniform values, the benchmark is $b(v) = 3v/4$. Gain and loss frames use identical numerical payoffs and differ only in wording. Our Space (https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning) structures the decision, self-reflection, and four-person peer play without storing responses. Higher loss-frame bids would qualify the benchmark; similar bids near $0.75v$ would retain it; departures in both frames would motivate revision of the bid rule or interface. Risk attitude, misunderstanding, and beliefs about opponents remain competing explanations. No

Table 1: Development roadmap.

Horizon	Milestone
Foundation	Bayesian games and auction design.
PS2	Conditional welfare test, 1,000 matched batches, and linked artifacts.
Symposium	Present the bounded claim and record peer challenges.
Next study	Framing experiment, human-bid import, and score audit.
2056	Accountable allocation of scarce AI capacity under private demand and verifiable risk.

human round has been completed, so behavioral evidence remains planned and exploratory.

5 Advanced development, real-world impact, and roadmap

The project builds on Harsanyi’s Bayesian games and Vickrey’s auction analysis. Its bounded contribution is a transparent comparison for scarce internal AI research with private PM values and a public, pre-bid current-risk score. In an earnings-season deployment, PMs, risk teams, clients, compliance staff, and computing operations are affected. A reserve may improve risk coverage and transparency, but score gaming, biased inputs, credit concentration, AI overreliance, and data leakage remain risks. Evidence of value requires a shadow deployment with frozen requests, score audits, decision logs, and pre-specified access, payment, override, and appeal outcomes. The next test combines a pre-registered framing study with a shadow allocation audit.

Expected 2056 Nobel Prize and Turing Award contribution statement. We aspire to develop validated theory and auditable infrastructure for allocating scarce AI capacity under private demand and verifiable risk, supported by formal results, replication, field evidence, and correction mechanisms.

Open Science Statement. The canonical computational repository is https://github.com/dku-comsci-econ206-Autumn2026/PS2_Qianshuo_Zhenning at final tested commit [59b1ead]. The runnable Colab notebook is available at https://colab.research.google.com/drive/1I-J52wxGaAIwJK_LjtghBnXS2GxY0U2V#scrollTo=QI1TUTULTNPj. The public Hugging Face Space is https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning. Primary computational results remain the executed Python outputs documented in Appendix A.2.

Statement of Contribution to the UN’s SDGs. The intended link is SDG Target 16.6 through an inspectable allocation rule. Decision records, eligible access, overrides, and appeals are proposed indicators; score and subgroup audits address possible bias. This is intended, not demonstrated, impact.

Author Notes

Acknowledgements

We thank **Prof. Luyao Zhang** for guiding the development of the project from its strategic research question through its game-theoretic, computational, behavioral, and global-leadership components, and for her formal course feedback. We thank **Prof. Ken Rogerson** and **Han Zhang** for their discussions and feedback that contributed to the development of the project.

We thank **Temur Akhtamjonov** and **Gihun Lee** for their peer reviews, particularly their comments on poster communication, figure presentation, and interpretation of the welfare threshold. We also thank **Longfei Jing** and **Jiayang Sun** for identifying the limitation of the assumed multi-slot bidding rule and for raising questions about endogenous and human bidding behavior. These comments led us to further qualify the computational claims, document sensitivity analysis, and separate completed simulation evidence from planned behavioral evidence.

We thank the **COMSCI/ECON 206 classroom community** for questions, discussion, and collaborative feedback during the symposium. We also acknowledge Python 3.9.6, Hugging Face Spaces, and OpenAI ChatGPT/Codex as described separately in Appendix A.3. The authors independently reviewed all model assumptions, code outputs, interpretations, and revisions and remain responsible for any remaining errors.

Feedback, classroom participation, or a program role does not automatically imply coauthorship. The authors remain responsible for the proposal, linked artifacts, and remaining errors.

Team and individual contributions

Joint intellectual contribution. Zhenning Wang and Qianshuo (Aaron) Wang jointly developed the research question of when a verified-risk reserve can improve the allocation of scarce AI research capacity relative to a pure first-price auction. The team specified the actors, private values, public risk scores, timing, allocation and payment rules, and the firm objective $W = V + \lambda C$. We jointly narrowed the contribution from a broad claim about AI-quota markets to an application-level comparison of FCFS, pure first-price auction, and a hybrid reserve mechanism. We also agreed to distinguish current risk coverage from avoided financial loss, internal payments from firm welfare, and completed computational evidence from planned behavioral evidence. The resulting proposal connects the analytical condition $\lambda \Delta C > L$, matched simulation results, the Hugging Face decision interface, and the planned gain/loss framing test.

Zhenning Wang. Zhenning Wang’s principal intellectual decision was to preserve the boundary between the completed computational comparison and the planned behavioral evidence, and to express that boundary through an inspectable decision interface. The resulting output is the public Hugging Face Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning, together with Sections 4–5 and the behavior-and-impact components of the poster. Zhenning verified that the Space uses the same $N = 20$, $K = 4$, $R = 1$, $r^* = 60$, assumed bid rule, mechanism sequence, and welfare accounting as the proposal. The Space also labels self-reflection and peer play as exploratory rather than population-level evidence. Following symposium feedback,

Zhenning recorded the behavioral and practical-impact comments, classified them as Keep, Revise, or Investigate, and checked the Space, paper, and poster for consistency.

Qianshuo (Aaron) Wang. Qianshuo Wang’s principal intellectual decision was to formulate the mechanism comparison as the measurable trade-off $\lambda \Delta C > L$ and to separate the single-slot equilibrium benchmark from the assumed multi-slot simulation bids. His inspectable output is the computational repository at https://github.com/dku-comsci-econ206-Autumn2026/PS2_Qianshuo_Zhenning, including the simulation, runnable notebook, verification script, and archived outputs. Six tests and a clean rerun reproduced the reported output files and primary mechanism-level results. This work connects the strategic model in Sections 1–2 to the computational answer in Section 3 and provides the rational benchmark used in the behavioral artifact. Following symposium feedback, Qianshuo recorded the behavioral and practical-impact comments, classified them as Keep, Revise, or Investigate, and checked the Space, paper, and poster for consistency.

References

- [1] John C. Harsanyi. 1968. Games with Incomplete Information Played by Bayesian Players, II: Bayesian Equilibrium Points. *Management Science* 14, 5 (1968), 320–334. doi:10.1287/mnsc.14.5.320
- [2] Parag A. Pathak, Tayfun Sönmez, M. Utku Ünver, and M. Bumin Yenmez. 2023. Fair Allocation of Vaccines, Ventilators and Antiviral Treatments: Leaving No Ethical Value Behind in Health Care Rationing. doi:10.48550/arXiv.2008.00374 arXiv:2008.00374v3, revised July 1, 2023.
- [3] Jinfei Sheng, Zheng Sun, Baozhong Yang, and Alan L. Zhang. 2026. Generative AI and Asset Management. *The Review of Financial Studies* (2026). doi:10.1093/rfs/hhag050 Advance article hhag050.
- [4] Balasubramanian Sivan, Renato Paes Leme, Mihai Tiuca, Ian McFarlane, Vasilis Gkatzelis, Nehal Mehta, Soheil Hassas Yeganeh, Vahab Mirrokni, and Amin Vahdat. 2026. Quota Marketplace: Dynamic Pricing for Efficient Allocation of ML Training Resources. doi:10.48550/arXiv.2607.09802 arXiv:2607.09802v1.
- [5] Murray Stokely, Jim Winget, Ed Keyes, Carrie Grimes, and Benjamin Yolken. 2009. Using a Market Economy to Provision Compute Resources Across Planet-wide Clusters. In *2009 IEEE International Symposium on Parallel & Distributed Processing*. IEEE, 1–8. <https://research.google/pubs/using-a-market-economy-to-provision-compute-resources-across-planet-wide-clusters/>
- [6] T. Rowe Price. 2026. T. Rowe Price Works With Anthropic To Bring Claude To More Of Its Investment Process. Press release, September 10. <https://www.troweprice.com/en/us/press/2026/press-release--t--rowe-price-works-with-anthropic-to-bring-claud> Accessed September 27, 2026.
- [7] William Vickrey. 1961. Counterspeculation, Auctions, and Competitive Sealed Tenders. *The Journal of Finance* 16, 1 (1961), 8–37. doi:10.1111/j.1540-6261.1961.tb02789.x

A Technical details and reproducibility

Model checks and the completed synthetic experiment are recorded below; the behavioral Space and poster are complete, while human behavioral evidence remains planned.

A.1 Model definitions and analytical checks

The synthetic scoring rule is $r_i = 100E_iS_i$, with normalized exposure E_i and downside severity S_i in $[0, 1]$. This is a policy score, not estimated research benefit. Freeze portfolios and requests; exclude duplicate requests, resale, and within-batch score manipulation. Credits have unit marginal opportunity cost. Correlated financial outcomes and endogenous portfolio choice are outside the baseline.

For the independent, uniform, single-slot benchmark, a type- v bidder mimicking type z against $\beta(z) = (n-1)z/n$ has expected utility $U(z; v) = [v - (n-1)z/n]z^{n-1}$. Its derivative is $(n-1)z^{n-2}(v-z)$,

maximized at $z = v$. Bids above the maximum equilibrium bid only raise payment; abstention gives zero. This verifies only the separate single-slot first-price benchmark used for theoretical and behavioral comparison; it is not claimed to be the exact equilibrium of the multi-slot hybrid mechanism. In the independent baseline, public risk scores do not update beliefs about private values because v_i and r_i are generated independently. Correlated value-risk environments are examined only in sensitivity analysis.

For any realized allocations, $W(H) - W(P) = -L + \lambda \Delta C$. If $L, \Delta C > 0$, improvement requires $\lambda > L/\Delta C$; if $L \geq 0$ and $\Delta C \leq 0$, strict improvement is impossible. As an arithmetic illustration, replacing $(v, r) = (90, 20)$ with $(60, 90)$ gives $L = 30$, $\Delta C = 70$, and threshold $3/7$. At $\lambda = 0.5$ the gain is 5; at $\lambda = 0.2$ it is -16 . This is not a simulation result. A pure first-price auction maximizes private value only when bids rank values correctly.

A.2 Computational inputs, algorithm, and verification

The fixed synthetic design uses $N = 20$, $K = 4$, $M = 100$, $R = 1$, $r^* = 60$, and $\lambda = 1$. Independently draw $v \sim U[0, 100]$ and exposure/severity $E, S \sim U[0, 1]$; freeze $r = 100ES$ before bidding. Use an independent random arrival permutation. The assumed bid $b = 0.8v$ is identical across mechanisms, including after the reserve outcome; it does not solve the multi-slot equilibrium. Budgets are nonbinding under these bids. Because every PM begins with the same 100-credit endowment, including unused credits gives total payoff $100 + x_i v_i - p_i$. The constant 100 does not affect strategic comparisons, so we report normalized utility

$$u_i = x_i v_i - p_i.$$

. Repeated draws are independent Monte Carlo replications, not strategic rounds.

Language-independent pseudocode. For each batch, freeze values, scores, arrival, and assumed bids. Pure first-price auction selects the top four bids and charges each its own bid. Hybrid selects up to R highest-risk eligible requests at zero payment, removes those PMs, then fills remaining capacity by first-price bids; unused reserves return to the auction. FCFS selects four by arrival and charges zero. Lower PM ID breaks all exact ties. Stop after allocation and payment; record each winner once. Compute paired differences on the same requests, then average across batches.

Metrics. Record winner IDs and total payments, V, C, W , and total utility $V - \sum p_i$. For the predetermined first batch, also report all individual payments and utilities. Value efficiency divides V by the top-four value sum. Eligible access divides eligible winners by eligible requests; no-eligible batches are NA and excluded from its mean. These metrics are distinct: risk coverage does not estimate avoided losses.

Fresh run. On September 26, 2026, Python 3.9.6 generated 1,000 primary batches with seed 20260926. No third-party packages are required. Run `python3 simulation.py`; `python3 verify_reproduction.py` executes six tests and reproduces 31,265 CSV cells in a clean temporary directory with numerical tolerance 10^{-10} . The executed notebook runs the same tests and simulation. Repository revision 59b1ead contains `fresh_run.json` fingerprints, `verification.txt`,

and complete CSVs in `reference_outputs.zip`. Tests cover reserve fallback, known displacement and payments, an already-winning reserve recipient, ties/eligibility boundary, budget/duplicate handling, and allocation/accounting invariants.

Sensitivity and limits. The 36 scenarios cross $r^* \in \{40, 60, 80\}$, $R \in \{1, 2\}$, three value-risk alignments, and two bid rules. Alignment stress tests sort the same values with or against risk; heterogeneous shading uses independent $b_i = \alpha_i v_i$, $\alpha_i \sim U[0.5, 1]$, fixed across mechanisms. Welfare weights are 0, 0.5, 1, 2. At the primary threshold/reserve, common shading gives mean $\Delta W = 0$ under positive rank alignment and -3.24 under negative rank alignment; heterogeneous shading gives $+13.12$ in the independent case. These are hypothetical behaviors, not peer observations. Under the primary design, only 43.8% of batches strictly improve W at $\lambda = 1$; the positive mean is not dominance. The paired Monte Carlo SE is 0.89, measuring simulation sampling uncertainty only. The break-even weight 0.733 is a ratio of mean loss to mean coverage gain, not a mean of batch thresholds. The design excludes strategic score manipulation, common values, endogenous arrival, and future decisions. Human-bid import and Hugging Face parity tests remain planned.

A.3 AI-use disclosure

OpenAI ChatGPT and Codex were used during September 2026 to support model clarification, literature checking, LaTeX editing, simulation implementation, debugging, and artifact drafting.

ChatGPT materially assisted with clarifying the distinction between the single-slot Bayesian benchmark and the stylized multi-slot bidding rule, refining the reserve-allocation sequence, separating internal payments from firm-wide welfare, and narrowing the novelty claim. The authors reviewed these suggestions and retained only those consistent with the course requirements and implemented model.

Codex assisted with reviewing draft sections, checking model and literature consistency, implementing and debugging the synthetic simulation, generating reproducibility checks, and reproducing archived outputs. Six automated tests and a clean rerun were used to verify the reported computational results.

No AI-generated output was accepted as evidence without author review. The authors independently selected the final research question, mechanism, parameterization, welfare definition, and behavioral design, and remain responsible for all claims, citations, computations, and limitations.

B Cumulative Intellectual Development

The project began as a general internal-auction problem in which portfolio managers competed for scarce AI research capacity using private bids. This initial framing revealed demand but did not clearly distinguish the project from existing resource-allocation applications.

During team development in September 2026, we revised the model by introducing a separate risk-reserve channel based on an auditable, bid-independent downside-risk score. This changed the research question from whether auctions can allocate scarce AI resources to when private willingness to pay should be supplemented by independently verified risk protection.

A second revision separated the exact single-slot first-price Bayesian benchmark from the multi-slot computational experiment. The equilibrium formula $b(v) = (n - 1)v/n$ is now used only for the single-slot theoretical and behavioral benchmark, while the multi-slot simulation explicitly labels $b_i = 0.8v_i$ as an assumed monotone bidding rule.

A third revision separated internal payments from firm-wide welfare and defined the main institutional trade-off as

$$W(H) - W(P) = -L + \lambda \Delta C.$$

The resulting PS2 contribution therefore shifted from applying an auction to a new AI setting toward testing when a hybrid auction-plus-reserve mechanism can improve the balance between private-value efficiency and verified downside-risk coverage.

C Open-source and three-lens inspiration

Software and source record. The completed computational experiment uses Python 3.9.6 and its standard library (official source: <https://www.python.org/>; PSF License). This choice supports a lightweight and reproducible simulation without third-party packages, but it does not validate the economic assumptions or establish external validity.

The behavioral artifact is hosted as a static Hugging Face Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning. It uses browser-side HTML, CSS, and JavaScript to structure the decision setting, benchmark comparison, reflection, and peer play. The Space is an interaction interface rather than a source of population-level evidence; the archived computational outputs remain the source of the reported 1,000-batch mechanism results.

The canonical computational repository is maintained at https://github.com/dku-comsci-econ206-Autumn2026/PS2_Qianshuo_Zhenning, at final repository commit 59b1ead. The runnable notebook and reproduction files document the implemented model, parameters, outputs, and limitations.

The runnable Colab notebook is available at https://colab.research.google.com/drive/1I-J52wxGaIwJK_LjtghBnXS2GxY0U2V#scrollTo=Q11TUTULTNPj. The corresponding notebook file is archived in the canonical repository as `PS2_comparison.ipynb`.

The interdisciplinary artifact is an A0 landscape poster that integrates the shared research question, verified gap, strategic model, three lenses, computational comparison, behavioral evidence status, real-world case, limitations, and links to the GitHub and Hugging Face artifacts. The editable poster file is `FP8_PS2_Poster.pptx`, with the current Canva version available at <https://canva.link/vlx5ljlj23z7qk8>.

The poster uses the same research question, baseline parameters, numerical results, mechanism definitions, limitations, instructor information, acknowledgements, and artifact links as the paper. All analytical charts, tables, and mechanism diagrams are team-generated from the reported synthetic results. The Duke Kunshan University logo is an external institutional visual.

Intellectual inspirations. Harsanyi’s Bayesian-games framework motivates the incomplete-information solution concept used for the single-slot bidding benchmark, while Vickrey’s auction analysis motivates strategic bid shading. Work on internal compute markets and quota-market allocation motivates treating high-capability

AI access as a scarce allocable resource, while reserve systems motivate protected non-price access. Related GenAI asset-management work establishes the application context. These sources motivate components of the design but do not directly answer when a verified current-risk reserve should displace private-value allocation in portfolio research.

Three-lens consequence. *Game theory* changes the prediction: private values and simultaneous bidding generate strategic bid shading rather than truthful reporting, with the exact equilibrium benchmark restricted to the separate single-slot case.

Social choice changes the evaluation: allocation quality is judged not only by private research value V but also by independently verified downside-risk coverage C , making the welfare weight λ explicit.

Mechanism design changes access: FCFS ignores both value and risk, the pure first-price auction allocates all capacity through bids, whereas the hybrid first-price auction with a risk reserve first protects one high-risk eligible request and then allocates the remaining capacity through first-price bidding.

Together, the three lenses change the prediction, evaluation criterion, and allocation rule rather than applying three labels to an unchanged mechanism.

D Review response and revision record

Review-ready version. The September 27 review-ready version uses the portfolio-manager AI research allocation model with $N = 20$, $K = 4$, one verified risk-reserve slot, and three comparison rules: FCFS, a pure first-price auction, and a hybrid first-price auction with one risk reserve. The computational baseline, sensitivity checks, artifact links, and current evidence status are frozen for review.

Pre-symposium revisions. Before the review-ready submission, the team revised the project in three material ways. First, we narrowed the contribution from a generic AI-quota auction to a portfolio-research setting in which private willingness to pay may not fully reflect verified downside risk. Second, we separated the exact single-slot first-price Bayesian benchmark from the assumed multi-slot bidding rule used in simulation. Third, we separated internal payments from firm-wide welfare and made the main trade-off explicit as private-value efficiency versus verified downside-risk coverage.

D.1 Formal response to September 28 feedback

D.1.1 Poster readability and visual density. Source and point raised. **Temur Akhtamjonov** noted that the Hugging Face game design was a strength, but that the A0 poster was visually crowded and some elements appeared too compressed. **Gihun Lee** similarly found the poster and repository well organized but indicated that some figures and tables would benefit from clearer presentation and captions.

Decision and reason: Revise. We accepted this feedback because the concern affects whether the strategic model, mechanism comparison, and computational evidence can be understood quickly during a poster presentation, even though it does not change the underlying model or results.

Change made. We reduced text density in the poster, increased spacing between major sections, shortened explanatory text around

figures, and revised figure and table captions so that the displayed comparison can be interpreted without relying on the oral presentation. We retained the same model assumptions, mechanisms, parameters, and numerical results.

Evidence location. See the revised A0 poster, especially the mechanism-comparison and computational-results panels. The underlying computational evidence remains unchanged and is reproduced in Section 3 and Appendix A.2 of the PS2 PDF, and in the GitHub repository at commit 59b1ead.

Remaining question. Poster readability has been revised, but further visual simplification could be tested with readers unfamiliar with the project. We treat this as a communication improvement rather than a change to the research design.

D.1.2 Interpretation of the break-even welfare weight. Source and point raised. Gihun Lee asked how the reported break-even value of approximately $\lambda = 0.733$ should be interpreted in a real asset-management setting.

Decision and reason: Revise. We accepted this concern because the original presentation could be misread as implying that 0.733 is an empirically estimated firm preference. It is instead a model-implied threshold under our synthetic calibration.

Change made. We clarified that the break-even welfare weight is calculated from the simulated mean private-value loss and mean risk-coverage gain,

$$\lambda^* = \frac{L}{\Delta C},$$

and therefore indicates how much weight the stated welfare function must place on one unit of verified risk coverage, relative to one unit of private research value, for the hybrid mechanism to break even against the pure auction. The value 0.733 is not an estimate of any real firm’s preferences and should not be interpreted as a directly observed policy parameter.

Evidence location. The interpretation is stated in Section 3 and Appendix A.2. The primary simulation gives a mean private-value loss of 26.82, a mean risk-coverage gain of 36.59, and a corresponding break-even ratio of approximately 0.733.

Remaining question. A real deployment would require the firm to justify or estimate an operational welfare weight using institutional objectives or observed decision outcomes. Estimating such a parameter from real organizational data remains planned future work.

D.1.3 Assumed bidding rule and endogenous strategic response.

Source and point raised. Longfei Jing observed that the trade-off between private value and risk coverage was clearly quantified, but noted that the computational bidding rule $b_i = 0.8v_i$ is assumed rather than derived from equilibrium. She asked whether the hybrid mechanism would still outperform the pure auction if bidders changed their strategies in response to the reserve rule.

Decision and reason: Investigate. We agree that endogenous strategic response is an important limitation. However, solving the exact equilibrium of the multi-slot hybrid mechanism would substantially expand the scope of the current PS2. We therefore do not reinterpret the assumed multi-slot bidding rule as an equilibrium result. Instead, we narrow the claim and test whether the welfare comparison is robust to alternative stylized bidding behavior.

Change made. We explicitly separate the exact single-slot first-price Bayesian Nash equilibrium benchmark,

$$b(v) = \frac{n-1}{n}v,$$

from the multi-slot simulation, where $b_i = 0.8v_i$ is only an assumed monotone bidding rule. We also report a heterogeneous-shading sensitivity condition,

$$b_i = \alpha_i v_i, \quad \alpha_i \sim U[0.5, 1],$$

held fixed across mechanisms.

Evidence location. The distinction between the equilibrium benchmark and the assumed simulation rule appears in Section 2, Section 3, Appendix A.1, and Appendix A.2. The heterogeneous-shading sensitivity results are reported in Appendix A.2 and reproduced in the GitHub simulation outputs at commit 59b1ead.

Remaining question. The current project does not solve the endogenous multi-slot equilibrium of the hybrid mechanism. Whether reserve eligibility changes equilibrium bidding strategies, and whether such strategic adaptation changes the welfare ranking, remains planned future work.

D.1.4 Behavioral validity of the assumed bid rule. Source and point raised. Jiayang Sun similarly noted that the welfare comparison depends on the assumed bidding rule and asked whether real participants would bid according to the modeled relationship between value and bid. She suggested running the Hugging Face behavioral interaction with human participants and comparing observed bids with the theoretical benchmark.

Decision and reason: Revise and Investigate. We accepted the interpretation concern and retained the behavioral test as an unresolved empirical question. Because no completed human-bidding round was available at the time of the final computational comparison, we do not present simulated bids as observed behavioral evidence.

Change made. We revised the paper and behavioral artifact to label $b_i = 0.8v_i$ as a computational assumption rather than an observed behavioral rule. The Hugging Face Space retains a separate four-bidder, single-slot task in which participants can submit bids under matched gain and loss frames and compare their choices with the rational benchmark $b(v) = 3v/4$.

Evidence location. See Section 4 for the planned behavioral comparison and Appendix F.1 for the behavioral protocol. The Hugging Face Space is: https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning.

Remaining question. Human bidding evidence has not yet established whether participants systematically follow, overbid, or underbid the rational benchmark, or whether gain/loss framing changes bids while numerical payoffs remain fixed. This remains planned exploratory work and is not used to support the completed welfare result.

E Structured abstract and cumulative proposal map

Proposal element	Current statement / change
Background	Internal compute markets and reserve systems already exist. This project examines whether a verified risk reserve can improve the allocation of scarce AI research capacity at an asset manager relative to a pure first-price auction.
Motivation and application	During earnings season, portfolio managers may bid according to private research value even when some requests carry broader verified downside risk for exposed portfolios. Competing PMs, the firm risk team, management, and clients are affected.
Three connected questions	Q1 asks when protecting verified risk justifies displaced private value. Q2 compares FCFS, a pure first-price auction, and a hybrid first-price auction with a risk reserve on matched computational batches. Q3 tests whether human bidding departs from the rational single-slot benchmark under fixed-payoff gain/loss framing. The shared question requires all three answers.
Method and model assumptions	Twenty unit-demand PMs compete for four premium AI research slots. Each PM has a private research value, while the firm observes a public, frozen, bid-independent risk score. In the hybrid mechanism, one eligible high-risk request receives the reserve slot before the remaining capacity is allocated through first-price bidding. The exact Bayesian Nash equilibrium benchmark is used only for the separate single-slot case; the multi-slot simulation uses an assumed monotone bid rule $b_i = 0.8v_i$.
Results or expected results	Analytically, the hybrid improves the stated welfare index when $\lambda\Delta C > L$. In the 1,000-batch synthetic comparison, the hybrid gains 36.59 units of risk coverage while losing 26.82 units of private value relative to the pure first-price auction. Mean $\Delta W = 9.78$ at $\lambda = 1$ and -8.52 at $\lambda = 0.5$, with a mean break-even welfare weight of approximately 0.733. Behavioral evidence remains planned until actual peer-play observations are collected.
Intellectual merits	The project does not claim novelty for auctions, reserve systems, or AI resource markets separately. Its contribution is a bounded application-level comparison that combines private strategic bids with an independent, verifiable risk-protection channel and makes the cost of displaced private value explicit.
Practical impacts	The proposed mechanism may help asset managers make explicit the trade-off between private research demand and verified downside-risk protection. Potential beneficiaries include portfolio managers, risk teams, management, and clients. Risks include mismeasured risk scores, systematic access bias, strategic gaming outside the modeled batch, and overreliance on synthetic calibration. Real-world use would require audited risk inputs, transparent eligibility rules, access monitoring, and evidence linking AI research priority to actual decision quality or downside outcomes.
Feedback received and revision made	Before symposium review, the team revised the project from a generic AI-quota auction to a portfolio-research mechanism with a verified risk reserve, separated the exact single-slot Bayesian benchmark from the assumed multi-slot bidding rule, and treated internal payments as transfers rather than firm-wide welfare losses. September 28 peer feedback identified three main revision priorities: poster readability, interpretation of the break-even welfare threshold, and the limitation of the assumed multi-slot bidding rule. The final revision clarifies these points, preserves the existing verified simulation, and records endogenous and human bidding as planned extensions.

Table 2: Appendix E structured abstract and cumulative proposal map.

F PS2 extension record: auction, linked artifacts, poster, and symposium

F.1 Auction laboratory record

The completed computational auction uses the inputs, allocation rules, payments, and outputs reported in Appendix A.2. The separate human behavioral round has not yet been conducted. Its planned identifier is HF-FP8-single-slot-v1. Four participants compete for one slot under an assigned gain or loss frame. Each participant receives a private value in $[0, 100]$ and submits one sealed bid in $[0, 100]$. The highest bidder wins, pays their own bid, and the game stops. The rational benchmark is $b(v) = 3v/4$.

The Hugging Face template records the assigned frame, private value, actual bid, benchmark deviation, winner, payment, utility, revenue, allocation efficiency, confidence, reflection, and resulting Keep/Qualify/Revise decision.

F.2 Linked-artifact parity

Table 3: Linked-artifact parity record.

Artifact	Parity check
GitHub	Commit 59b1ead; model, executed notebook, outputs, and clean rerun are recorded in Appendix A.2. https://github.com/dku-comsci-econ206-Autumn2026/PS2_Qianshuo_Zhenning
Hugging Face	Live static Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Qianshuo_Zhenning . It uses the same decision setting, three mechanisms, behavioral benchmark, planned evidence status, and privacy boundary as the proposal.
A0 poster	Editable poster file: FP8_PS2_Poster.pptx. Current Canva version: https://canva.link/vlx5ljli23z7qk8 . The poster uses the same research question, mechanism definitions, baseline numerical results, real-world case, limitations, instructor information, acknowledgements, artifact links, and selected references as the paper.

F.3 Symposium and practical-impact record

The review-ready claim for the symposium is that, under the stated primary parameters, the hybrid mechanism increases mean risk coverage enough to produce a positive mean ΔW at $\lambda = 1$. This is not universal dominance: the result depends on the welfare weight, value-risk alignment, risk-score validity, and assumed bids. The real-world case therefore requires score audits, decision records, access monitoring, and a correction route.

Symposium feedback and final revision. The September 28 peer reviews raised three main issues: poster readability, interpretation of the break-even welfare weight, and the reliance of the multi-slot simulation on an assumed bidding rule. **Temur Akhtamjonov** and **Gihun Lee** motivated revisions to poster presentation and explanation of the welfare threshold. **Longfei Jing** and **Jiayang Sun** questioned whether the assumed rule $b_i = 0.8v_i$ would remain appropriate under endogenous or human bidding.

The team classified the presentation and interpretation issues as *Revise*, and the endogenous-bidding question as *Investigate*. We

clarified that $\lambda = 0.733$ is a synthetic break-even threshold rather than a real-world estimated preference parameter, strengthened the distinction between the exact single-slot BNE benchmark and the assumed multi-slot bidding rule, and retained heterogeneous bid shading as a robustness check. Human bidding under the Hugging Face protocol remains planned exploratory work.

No symposium comment changed the implemented baseline parameters or primary mechanism definitions. Therefore, the primary simulation was not replaced; the existing clean rerun and sensitivity analysis remain the verified computational evidence. The final PDF, poster, GitHub repository, and Hugging Face Space were checked for consistent mechanism definitions, evidence labels, numerical claims, and limitations.